

Supervised Learning

Model Building & Evaluation Summary Report

Datasets: Housing Prices & Titanic Survival

Analyst Lab Africa Data Science Internship Program

Tools used: Python · Pandas · NumPy · Scikit-learn · Seaborn · Matplotlib

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1. Introduction

This report documents the supervised learning models built on the two datasets used throughout this internship: the Housing Prices dataset and the Titanic Survival dataset, both sourced from Kaggle. Building on the data cleaning, exploratory data analysis, and statistical testing completed in previous weeks, this stage moves from understanding the data to actually building predictive models.

The Housing dataset required a regression approach, since the target variable — price — is a continuous number. The Titanic dataset required a classification approach, since the target variable — survival — is a binary category (0 or 1). Feature selection for both models was guided directly by the correlation analysis, hypothesis testing, and causation testing carried out in the Statistics & Probability stage, rather than chosen arbitrarily.

2. Part A — Housing Dataset (Regression)

2.1 Dataset

Source: Housing Prices dataset, Kaggle.

The dataset contains 545 records of houses with features including area, bedrooms, bathrooms, stories, parking, and several binary amenities (airconditioning, mainroad, guestroom, basement, hotwaterheating, prefarea) plus furnishing status. The target variable is price.

2.2 Steps Taken

1. Selected features based on Week 3 correlation analysis, starting with the four strongest predictors: area (0.54), bathrooms (0.52), airconditioning (0.45), and stories (0.42).
2. Applied a log transformation to price as the prediction target, since price was confirmed right-skewed (skewness 1.21) and failed the Shapiro-Wilk normality test during Week 3.
3. Split the data into 80% training (436 houses) and 20% testing (109 houses) using `train_test_split` with `random_state=42` for reproducibility.
4. Trained a Linear Regression model on the training set and generated predictions on the held-out test set.
5. Evaluated performance using R^2 , RMSE, and MAE — both in log scale and converted back to actual naira values for interpretability.
6. Tested whether adding Bedrooms improved the model, despite its comparatively weaker correlation (0.37) and overlap with bathrooms and stories.
7. Tested whether adding Prefarea (0.33) and Parking (0.38) — both tied to location and access — improved on the four-feature baseline.
8. Visualized actual vs predicted price for each model version to inspect where the model performed well and where it struggled.

2.3 Model Performance

Model 1 — Baseline (Area, Bathrooms, Airconditioning, Stories)

Metric	Value
R ² Score	0.5470
RMSE (log scale)	0.2957
MAE (log scale)	0.2393
RMSE (actual price)	₦1,521,795.53
MAE (actual price)	₦1,140,088.39

This baseline model, trained on the four strongest predictors identified in the correlation analysis, explained about 55% of the variation in house prices, with a typical prediction error of roughly ₦1.52 million (RMSE). In practical terms, using only a house's area, bathroom count, airconditioning, and number of stories, the model could account for just over half of why prices differ across the dataset — the remaining 45% reflects factors such as exact location, parking, and furnishing that the model never saw.

RMSE punishes large errors more heavily than small ones, since it squares each error before averaging. A relatively high RMSE here signals that the model is not simply a little off everywhere — it is getting a handful of houses very wrong, and those large misses are pulling the metric up.

Feature Test — Adding Bedrooms

Model	R ²
4 features (no bedrooms)	0.5470
5 features (with bedrooms)	0.5579

Adding bedrooms improved R² by only 0.0109 (about 1.1 percentage points). Given that bedrooms is moderately correlated with both bathrooms (0.37) and stories (0.41) — features already in the model — this small gain is consistent with multicollinearity: bedrooms carries some genuinely new information, but most of what it would explain is already captured by the other features. The baseline four-feature model was kept for simplicity and interpretability, since the marginal gain did not justify added model complexity.

Model 2 — Adding Prefarea and Parking

Metric	Model 1 (4 features)	Model 2 (6 features)
R ² Score	0.5470	0.5942
RMSE (actual price)	₦1,521,795.53	₦1,425,286.51
MAE (actual price)	₦1,140,088.39	₦1,066,465.21

Adding Prefarea (preferred location) and Parking produced a more meaningful improvement than bedrooms did — R² rose from 55% to 59%, and RMSE dropped by about ₦96,500. This confirms that location desirability genuinely affects price beyond what physical size and amenities alone can explain. A house in a preferred area commands a premium that specifications like area and bathrooms cannot capture on their own.

However, the actual-vs-predicted scatter plot revealed that both models share the same structural weakness: houses priced above roughly ₦8 million are consistently under-predicted, in some cases by several million naira — for example, a house actually priced near ₦13.2 million was predicted at only about ₦9 million even in the improved six-feature model. Since this pattern persisted after adding two genuinely useful features, it points to a limitation of the model type itself rather than a missing feature: a linear model assumes every extra bathroom or storey adds a fixed amount to price regardless of how expensive the house already is, and that assumption appears to break down at the luxury end of the market.

2.4 Challenges Encountered

- Virtual environment and kernel mismatches caused repeated `ModuleNotFoundError` issues when packages were installed in one `.venv` while the notebook kernel pointed to a different Python environment.
- A pandas `ChainedAssignmentError` appeared when using the older `df['col'].fillna(value, inplace=True)` syntax, requiring a switch to the updated `df['col'] = df['col'].fillna(value)` pattern introduced with pandas' Copy-on-Write behaviour.
- An `IntCastingNaNError` occurred when converting the `Embarked` column to integer type before all missing values had been filled, requiring the fill step to run before the type conversion.
- A `house_id` column derived from a pre-sorted index initially produced a misleadingly strong negative correlation (-0.93) with price purely due to row ordering, and had to be identified and removed before correlation analysis could be trusted.
- `Fare` contained values of exactly 0, which broke a direct log transformation; this was resolved by transforming `log(Fare + 1)` instead.

2.5 Key Insights

- Area and Bathrooms remain the strongest individual predictors of price, consistent with the correlation and regression analysis from Week 3.
- Bedrooms added only a marginal improvement (+1.1% R^2) once Bathrooms and Stories were already included, confirming the multicollinearity identified earlier — extra bedrooms largely overlap with information the model already has.
- `Prefarea` and `Parking` provided a genuine, larger improvement (+4.7% R^2), confirming that location and access are real, independent drivers of price beyond physical specifications.
- Both the 4-feature and 6-feature linear models systematically under-predict luxury houses (above ~₦8 million), indicating that a straight-line model cannot fully capture the pricing behaviour of the most expensive properties in this dataset.
- This limitation suggests the next step should be a non-linear model, such as Random Forest, which can capture relationships that do not scale at a constant rate across the price range.

3. Part B — Titanic Dataset (Classification)

3.1 Dataset

Source: Titanic Survival dataset, Kaggle.

The dataset contains 891 passenger records. After cleaning, the working features included Pclass, Sex, Age, SibSp, Parch, Fare, and Embarked, with the target variable Survived (0 = died, 1 = survived).

3.2 Steps Taken

9. Selected features deliberately based on Week 3 findings rather than including every available column: Pclass (correlation -0.34) and Sex (correlation 0.54) were kept as strong, independent predictors confirmed by logistic regression; Age was kept despite a weak correlation (-0.07) because it was statistically significant in hypothesis testing and theoretically justified by the 'women and children first' policy.
10. Verified there were no remaining missing values before modelling, since Logistic Regression cannot handle nulls.
11. Split the data into 80% training (712 passengers) and 20% testing (179 passengers) using `train_test_split` with `random_state=42`.
12. Trained a baseline Logistic Regression model on Pclass, Sex, and Age, then generated predictions on the test set.
13. Evaluated performance using accuracy, a confusion matrix, and a full classification report (precision, recall, F1-score).
14. Tested whether adding Fare improved the model, despite Week 3's finding that Fare's effect on survival collapses once Pclass is controlled for.
15. Tested `class_weight='balanced'` on the baseline model to address the dataset's class imbalance (62% died vs 38% survived) and improve the model's ability to correctly identify survivors.

3.3 Model Performance

Model 1 — Baseline (Pclass, Sex, Age)

Metric	Value
Accuracy	81.01%
Precision (Survived)	0.79
Recall (Survived)	0.73
F1-score (Survived)	0.76
Precision (Died)	0.82
Recall (Died)	0.87

Confusion Matrix: 91 true negatives, 14 false positives, 20 false negatives, 54 true positives (out of 179 test passengers).

This model, trained using only three features, correctly predicted survival for 81% of passengers it had never seen before. In practical terms, if given the class, sex, and age of 100 strangers from the Titanic, the model would correctly guess survival for about 81 of them. This is a strong result given the model relies on just three pieces of information, and it reflects real, historically documented patterns: ticket class often determined cabin location and proximity to lifeboats, sex determined evacuation priority under the 'women and children first' policy, and age factored into who was prioritized for rescue. The remaining 19% of errors represent passengers whose individual outcome did not follow the broader statistical pattern.

Feature Test — Adding Fare

Metric	Baseline (no Fare)	With Fare added
Accuracy	81.01%	79.89%
Recall (Survived)	0.73	0.73
Precision (Survived)	0.79	0.77

Adding Fare slightly reduced accuracy rather than improving it. This directly confirms the Week 3 causation finding: in the multiple logistic regression run during the correlation-versus-causation analysis, Fare's coefficient collapsed to 0.0008 once Pclass was included, showing that Fare carries almost no independent information beyond what Pclass already captures. Reintroducing Fare here added redundant noise rather than new predictive signal, which is exactly why performance dipped slightly instead of improving.

Model 2 — Balanced Class Weights (Pclass, Sex, Age)

Metric	Baseline	Balanced
Accuracy	81.01%	80.45%
Recall (Survived)	0.73	0.85
Precision (Survived)	0.79	0.72
F1-score (Survived)	0.76	0.78

Confusion Matrix (balanced): 81 true negatives, 24 false positives, 11 false negatives, 63 true positives.

Applying `class_weight='balanced'` produced a large improvement in recall for survivors — from 73% to 85% — meaning the model now correctly identifies 63 of 74 actual survivors in the test set, missing only 11, compared to missing 20 in the baseline. This came at the cost of more false positives (24 versus 14) and a small drop in overall accuracy (80.45% versus 81.01%). This is a meaningful trade-off rather than a straightforward improvement: the balanced model is better suited to a context where missing a real survivor is more costly than a false alarm, while the baseline model is better suited to contexts where overall accuracy is the primary measure of success.

3.4 Challenges Encountered

- Initial feature selection defaulted to including all available columns (Pclass, Sex, Age, SibSp, Parch, Fare, Embarked) before being deliberately narrowed down to the three features already justified by Week 3's correlation and causation analysis.

- Confusion about why Logistic Regression, rather than Linear Regression, was the correct model — resolved by clarifying that Survived is a binary category, not a continuous number, and Linear Regression cannot constrain its output to a valid 0/1 probability range.
- The dataset's class imbalance (62% died, 38% survived) caused the baseline model to under-perform specifically on recall for survivors, which was only identified by inspecting the full classification report rather than relying on accuracy alone.

3.5 Key Insights

- A simple three-feature Logistic Regression model (Pclass, Sex, Age) achieves a strong baseline accuracy of 81%, confirming these as meaningful, independent predictors of survival.
 - Adding Fare back into the model slightly reduced performance, providing direct empirical confirmation of the Week 3 finding that Fare is a confounded proxy for Pclass rather than an independent cause of survival.
 - Accuracy alone can be misleading on an imbalanced dataset: the baseline model had the highest overall accuracy, but the balanced model caught significantly more real survivors (85% vs 73% recall), a meaningful difference depending on the real-world goal of the prediction.
 - The choice between the baseline and balanced model is a genuine trade-off, not a simple upgrade — the right choice depends on whether the priority is overall correctness or minimizing missed survivors.
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4. Overall Conclusion

Across both datasets, model performance closely tracked the statistical groundwork laid in the earlier stages of this internship. In the Housing dataset, the features with the strongest, most independent correlations (Area, Bathrooms, Prefarea, Parking) produced real improvements in R^2 and RMSE, while a feature with known overlap (Bedrooms) produced only a marginal gain — confirming that the correlation and multicollinearity analysis from Week 3 translated directly into modelling outcomes. In the Titanic dataset, the feature proven to be confounded by causation testing (Fare) measurably failed to improve the model when reintroduced, while the features confirmed as independent predictors (Pclass, Sex, Age) formed a strong, defensible baseline on their own.

Both models also revealed clear directions for further improvement: the Housing model's persistent under-prediction of luxury houses points toward the need for a non-linear approach such as Random Forest, while the Titanic model's imbalance-driven trade-off between accuracy and recall highlights the importance of selecting an evaluation metric that matches the real-world goal of the prediction, rather than defaulting to accuracy alone.